ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A5 · PLANT GROWTH

Seeds and plant needs

★☆☆
Grade 5

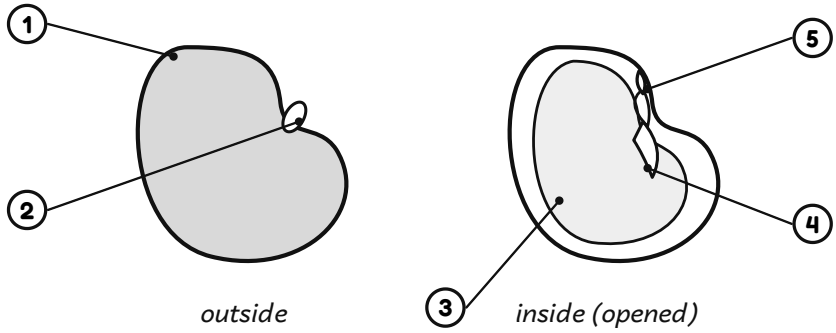
Learning goals

- Label the parts of a seed and a plant.
- Recall the conditions for germination and growth.

1 Label the parts of the bean seed. [5]

LABELS

seed coat **scar** **embryo root** **food store** **embryo shoot**



1 seed coat
2 scar
3 food store
4 embryo root
5 embryo shoot

2 Complete the sentences. Use the words in the box. [7]

WORD BANK

absorbs **warmth** **flower** **food store** **germinates** **soil** **root** **light** **shoot**

When a seed germinates, it first absorbs water and swells. The seed coat splits and the first root grows downwards. Next the shoot grows upwards. The seed gets the energy it needs from its food store. Seeds need water, warmth and air to germinate, but they do not need light.

3 Draw a line to match each plant part to its job. [4]

Draw a line to match each pair, or write the letter.

roots ●	→ A	● **A** take in water and hold the plant in the soil
stem ●	→ C	● **B** use light energy to make food
leaves ●	→ B	● **C** holds the plant up and carries water
flower ●	→ D	● **D** makes seeds for new plants

4 **True or false?** [5]

Statement	True	False	If false, correct it
1. Seeds need light to germinate.		✔	Seeds do not need light to germinate.
2. A plant kept in the dark grows pale and weak.	✔		
3. Chlorophyll is the green substance that traps light.	✔		
4. The embryo grows into the new plant.	✔		
5. Plants grow away from the light.		✔	Plants grow towards the light.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A5 · PLANT GROWTH

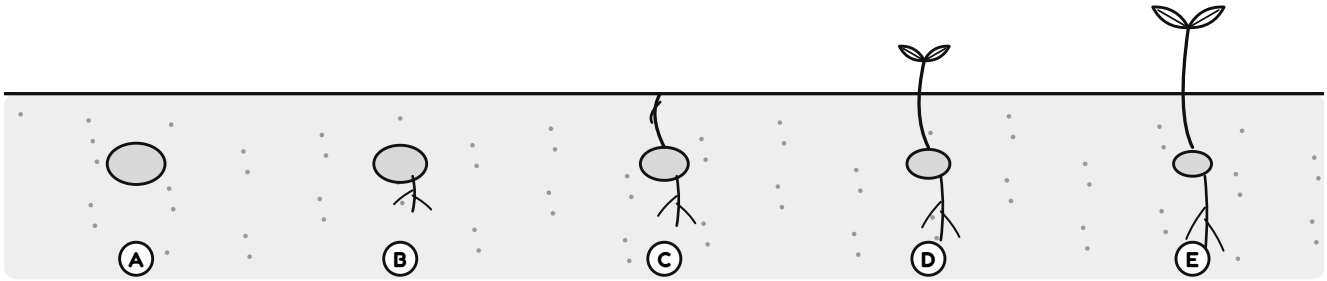
Explaining germination and growth

★★☆
Grade 5

Learning goals

- Order the stages of germination.
- Use results to decide what seeds need.
- Explain the effect of missing factors.

1 **Write the letter of each stage (A to E) next to its description.** [5]



Description	Stage
The seed has absorbed water and swollen.	A
The first root grows down.	B
The shoot pushes up through the soil.	C
The first leaves open above the ground.	D
The leaves are bigger; the seed has shrivelled.	E

2 **Class 5A put 20 cress seeds in each of four dishes. After 5 days they counted how many had germinated.** [6]

Dish	Conditions	Seeds germinated (out of 20)
1	warm, wet cotton wool, in the light	18
2	warm, wet cotton wool, in a dark cupboard	19
3	cold (fridge), wet cotton wool, dark	2
4	warm, dry cotton wool, in the light	0

a) Do seeds need light to germinate? Use dishes 1 and 2 to explain.

No. Dishes 1 (light) and 2 (dark) had almost the same number germinate (18 and 19).

b) Which dish shows that seeds need water?

Dish 4: with no water, no seeds germinated.

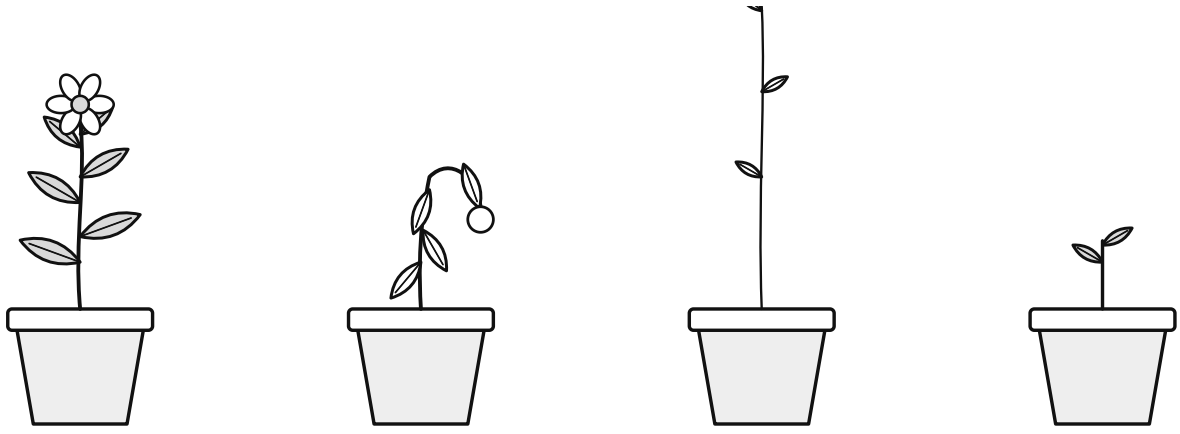
c) Which dish shows that seeds need warmth? Explain.

Dish 3: in the cold only 2 germinated compared with 19 in the warm dark dish 2.

d) Why did the class use 20 seeds in each dish, not just one?

One seed might be dead or damaged; more seeds give more reliable results.

3 **Look at the four plants. For each one, say which factor is missing and how you can tell.** [6]



light, water, warmth, air | no water | no light | too cold

Plant	Missing factor	What you can see
2	water	drooping (wilted) stem and leaves
3	light	tall, thin, pale, few leaves
4	warmth	very small, has hardly grown

4 **Why does *a* seed need *a* food store, but *a* grown plant does not?** [2]

A seed has no leaves yet, so it cannot make food from light; the food store gives energy until the leaves grow. A grown plant makes its own food in its leaves using light.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A5 · PLANT GROWTH

Does light affect plant growth?

★★☆
Grade 5

Learning goals

- Plan a fair test.
- Present results in a line graph.
- Use evidence to write a conclusion.

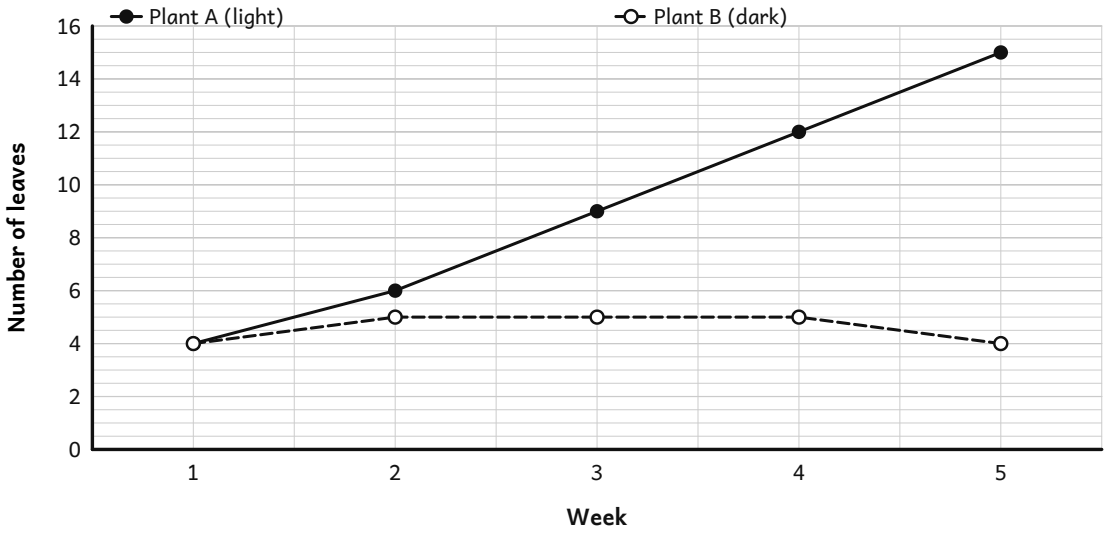
Nadia and Tanvir had two pots of the same kind of plant, each with 4 leaves. They put **Plant A** on a sunny windowsill and **Plant B** in a dark cupboard. Each week they gave both plants 50 ml of water and counted the leaves.

1 **Complete the plan for their fair test.** [5]

Question	Does light affect how well a plant grows?
What we change	light (windowsill or dark cupboard)
What we measure	the number of leaves each week
What we keep the same	type and size of plant, pot, soil, amount of water, temperature
Prediction	Plant A will grow more leaves because it gets light to make food.

2 **Here are their results. Draw a line graph with both plants on the same axes.** [5]

Week	1	2	3	4	5
Plant A: leaves (light)	4	6	9	12	15
Plant B: leaves (dark)	4	5	5	5	4



3 **Why is a line graph better than a bar chart for these results?** [1]

Line graphs show how something changes over time; here the number of leaves changes week by week.

4 **What happened to Plant B in week 5? Suggest why.** [2]

It lost a leaf (4 leaves). Without light it could not make food, so leaves turned yellow and fell off.

5 **Write a conclusion for the investigation.** [2]

Plants need light to grow well: the plant in the light grew many more leaves than the plant in the dark.

Remember

Measure from the same point each time, and repeat measurements to make your results more reliable.

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REAL-WORLD CASE STUDY · A5 · PLANT GROWTH

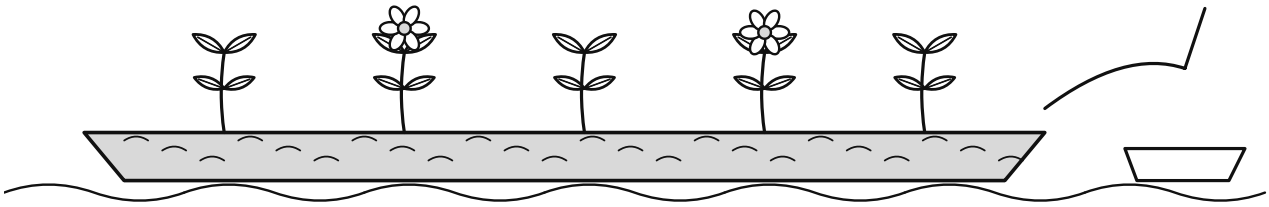
Floating gardens: farming on flood water

★★★
Grade 5

Learning goals

- Apply ideas about plant needs to a real farming method.
- Consider how people adapt to their environment.
- Link science to climate, culture and history.

Science · Geography · History · Climate change · Design



A floating garden made from water hyacinth, with vegetable seedlings growing on top.

In the south of Bangladesh, in districts such as Gopalganj, Pirojpur and Barishal, low land stays under water for several months during the monsoon. Farmers cannot plant crops in the flooded fields. For hundreds of years, families here have solved this problem with **floating gardens**, called *dhap* or *baira*.

Farmers pile up water hyacinth (a floating water plant), straw and rice stubble to make a thick raft, about one metre deep. As the plants slowly rot, they make a rich, soft compost full of **nutrients**. Farmers sow seeds of okra, gourds, red amaranth and other vegetables on top. The gardens rise and fall with the water, so the crops never drown. When the floods go down, the rotted rafts are spread on the fields as fertiliser.

In 2015 the Food and Agriculture Organization of the United Nations named Bangladesh's floating gardens a **Globally Important Agricultural Heritage System**. As climate change brings more floods, more farmers are interested in this old idea.

Sources: FAO GIAHS programme; descriptions of floating agriculture in southern Bangladesh.

1 **List the four factors that plants need to grow. Explain how a floating garden gives the plants each one.** [4]

Factor	How the floating garden provides it
water	the roots reach moisture from the raft and the water below
light	the garden floats in the open sunlight
air	the leaves and stems are above the water, not drowned
nutrients / warmth	the rotting water hyacinth makes compost full of nutrients; it is warm in the monsoon

2 **Why would seeds planted in a flooded field not germinate, even though there is plenty of water?** [2]

Seeds also need air (oxygen) to germinate; under flood water they cannot get enough air and they rot.

3 **Floating gardens use water hyacinth, a plant that grows so fast it blocks rivers. Why is this a clever idea?** [2]

It turns a problem weed into something useful (free material and compost) and clears waterways.

4 **Think globally. In Mexico, the Aztecs grew food on *chinampas*, gardens built in lakes. What do people in both places have in common?** [2]

Both used local materials and knowledge to grow food where land was covered by water: they adapted to their environment.

5 **Design. Draw a floating garden that could grow vegetables for a school. Label the materials and show how the plants get water, light, air and nutrients.** [4]

Your design

Accept any design with a floating base (e.g. hyacinth/straw raft or plastic bottles), soil or compost layer, plants on top, and labels for the four needs.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A5 · PLANT GROWTH

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

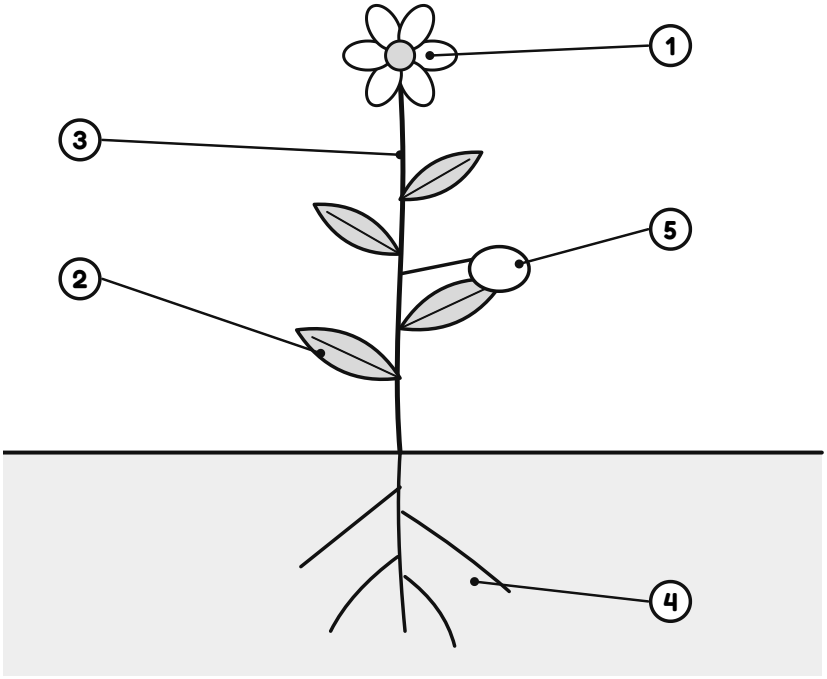
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F	I	F	D	U	A	S	E	E	D	E
A	B	W	B	O	C	G	M	J	U	A
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C	N	M	O	E	R	Y	L	S	E	O
S	M	T	W	T	F	W	E	T	M	O
Y	I	H	U	A	U	P	A	E	B	T
L	D	B	I	R	T	E	F	M	R	F
V	S	H	O	O	T	E	C	F	Y	H
A	G	F	P	O	T	S	R	S	O	M

embryo **factor** **leaf** **light** **root** **seed** **shoot** **stem** **warmth** **water**

2 **Label the parts of the plant.** [5]

LABELS

flower leaf roots stem fruit



1 flower

2 leaf

3 stem

4 roots

5 fruit

3 **Which part of a seed grows into the new plant?** [1]

- ○ seed coat
- ● **embryo**
- ○ food store
- ○ scar

4 **A seed is planted deep in dark soil. Will it germinate? Explain.** [2]

Yes, if it has water, warmth and air: seeds do not need light to germinate because they use their food store.

5 **Challenge. Shabnam says, "Plants get their food from the soil." Is she right? Explain.** [2]

Not quite. Plants take water and nutrients (minerals) from the soil, but they make their food in their leaves using light energy.